Computer Networks
CSA0712
Co-5-Assessment Tool-2

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1) SMTP Transaction with Explicit Error Handling

```
FUNCTION SecureSendEmail(sender, recipient, messageBody)
    Connect to SMTP server on Port 25

    Send "HELO"
    response ← Read Server Response
    StatusCode ← First 3 digits of response.

  IF StatusCode starts with 4 OR statusCode
Starts with 5 THEN
        SEND "QUIT"
        RETURN "Error: HELO command failed"
   END IF

   SEND "MAIL FROM: <sender>"
   response ← Read server Response
   StatusCode ← First 3 digits of response

   IF StatusCode state starts with 4 OR
StatusCode starts with 5 THEN
        SEND "QUIT"
        RETURN "ERROR: Mail FROM Command
 failed"
    END IF.
   SEND "RCPT TO: <recipient>"
   response ← Red Server Response
```

```
    statusCode ← First 3 digits of response.
  IF statusCode starts with 4 OR statuscode
starts with 5 THEN
      SEND "QUIT".
      RETURN "ERROR: RCPT TO command failed"
  END IF.
  SEND "DATA"
  response ← Read server Response
  statuscode ← First 3 digits of response
  If statuscode starts with 4 OR statuscode
starts with 5 THEN
      Send "QUIT"
      Return "Error: DATA command failed"
  END IF.
  SEND message Body
  SEnd "."
  response ← Read server Response
  Status Code ← First 3 digits of response.
  IF statusCode starts with 4 OR statusCode
starts with 5 THEN
     SEND "QUIT"
     RETURN "Error: Message transmission
failed"
   END IF
   SEnd "QUIT"
   Return "Email sent successfully".
END FUNCTION
```

2. Pop3 Selective Download and storage.

```
PROCEDURE SyncInboxPOP3 (username, password,
local Database)
    Connect to POP3 server on Port 110
    READ Server greeting
    Send "USER Username"
    Read server Response
    Send "PASS Password"
    Read Server Response
    Send "STAT"
    response ← Read Server Response
    total messages ← Extract Number of Mes-
sages from response
    FOR message Number ← 1 to total messags
        SEND "UIDL messagenumber"
        response ← Read Server Respons
        uniqueID ← Extract UIDL from respons
        IF local Database contains uniqueID
THEN Print "Email already exists.
ELSE
    SEnd "RETR MessageNumber"
    emailData ← Read complete Email
    Save emailData and uniqueID into
localdatabase.
```

```
        print "New email downloaded and saved."
    ENDIF
ENDFOR.
    Send "QUIT"
    Retuan "Inbox synchronization completed"
END PROCEDURE.
```

3. IMAP Dynamic Flag management.

```
PROCEDURE ArchiveOld Emails()
    Connect to IMAP server on port 143
    Send "LOGIN username password"
    Read server Response
    Send "SELECT INBOX"
    Read server Response.
    Send "SEARCH SEEN"
    respons ← Read server Response
    messageIDS ← Extract Message IDs from
response.
    FOR EACH Message ID IN MessageIDS.
        Send "COPY messageID OLDMAIL"
        response ← Read server Response
    IF COPY was successful THEN
        Send "STORE messageID+FLAGS
(\Deleted)"
        Read server Response
    ELSE
        PRINT "Copy failed for message" +
messageID.
```

```
    ENDIF
    ENDFOR
    Send "EXPUNGE"
    Read Server Response
    Send "LOGOUT"
    Return "old read emails archived successfully"
ENDPROCEDURE
```

4. Socket Timeout Recovery in Email protocols.

```
FUNCTION RobustNetworkReader(socket, timeout)
    retryCount ← 0
    WHILE retryCount < 3
        StartTime ← CurrentTime
        Try to read data from socket
        WHILE data has not been received
            currentTime ← CurrentTime
            elapsedTime ← CurrentTime - startTime
        IF elapsedTime >= timeoutLimit THEN
            retryCount ← retryCount + 1
            Print "Server response timed out"
            Print "Retry attempt": +
retry count
            IF retryCount < 3 THEN
                Retry reading from socket
            BREAK
```

```
        ELSE
            Close socket forcefully
            THROW
    Connection Timeout Exception
                ENDIF
            END IF
        END WHILE
        IF data was received THEN
            Return data
        END IF
    END WHILE
END FUNCTION
```